Agent Mesh - Sprint 1 Application Test Cases

Team 2 - Phoenix. Covers US_01 to US_05. All 23 test cases are automated and passed in CI on 2026-10-18 (CI workflow). Backend cases run in the Backend job; UI cases run in the Playwright e2e job against the Docker stack in Demo Mode.

TC_ID	US_ID	Test Scenario	Preconditions	Test Steps	Expected Result	Type	Automated Test	Status
TC_01	US_01	Enroll an agent with valid details	Backend running; agent name not used	1. Send an enrollment request with all required fields. 2. Open the registry.	Agent is created with a unique ID and Active status and appears in the registry.	Positive	backend/tests/test_sprint1_acceptance.py:: test_tc01_enroll_agent_with_valid_details	**Pass**
TC_02	US_01	Enroll with a missing required field	Backend running	1. Send an enrollment request without capabilities.	Request is rejected with a validation error naming the field; no agent is created.	Negative	backend/tests/test_sprint1_acceptance.py:: test_tc02_enroll_with_missing_required_field, test_tc02_empty_capability_list_is_rejected	**Pass**
TC_03	US_01	Enroll a duplicate agent name	An agent with this name exists	1. Enroll another agent with the same name.	Request is rejected with a duplicate-name error; the original agent is unchanged.	Negative	backend/tests/test_sprint1_acceptance.py:: test_tc03_enroll_duplicate_agent_name, test_tc03_builtin_names_are_also_protected	**Pass**
TC_04	US_01	Enrollment speed and single request	Backend running	1. Enroll 10 agents one after another and time each request.	Each enrollment finishes in a single request in under 3 seconds with no duplicates.	Positive	backend/tests/test_sprint1_acceptance.py:: test_tc04_enrollment_is_a_single_fast_request	**Pass**
TC_05	US_01	Disable an enrolled agent	An active agent exists	1. Disable the agent. 2. Submit a brief that would need it.	The disabled agent is not spawned and is not offered to the orchestrator.	Positive	backend/tests/test_sprint1_acceptance.py:: test_tc05_disabled_agent_is_never_spawned	**Pass**
TC_06	US_02	Brief needing several capabilities	Registry has at least 5 agents	1. Submit: audit this code, fix critical issues, write tests, produce a report.	Multiple agents are spawned, each with its own instance ID and status; unrelated agents are skipped.	Positive	backend/tests/test_sprint1_acceptance.py:: test_tc06_brief_needing_several_capabilities	**Pass**
TC_07	US_02	Brief needing one capability	Registry has at least 5 agents	1. Submit: clean this CSV file.	Only the data cleaner is spawned; all others are skipped with a reason.	Positive	backend/tests/test_sprint1_acceptance.py:: test_tc07_brief_needing_one_capability	**Pass**
TC_08	US_02	Brief with no matching agent	Registry has no agent for the capability	1. Submit a brief asking for a capability nobody offers.	No agent is spawned and the user is told which capability is missing.	Negative	backend/tests/test_sprint1_acceptance.py:: test_tc08_brief_with_no_matching_agent	**Pass**
TC_09	US_02	Empty or vague brief	Workspace open	1. Submit an empty brief. 2. Submit: help me.	Empty brief is blocked; vague brief gets a clarifying question; no agents are spawned.	Negative	backend/tests/test_sprint1_acceptance.py:: test_tc09_empty_brief_is_blocked, test_tc09_vague_brief_gets_a_clarifying_question	**Pass**
TC_10	US_02	Same brief submitted twice	Registry seeded	1. Submit the same brief two times.	Two separate sets of agent instances are created with different IDs and no data collision.	Positive	backend/tests/test_sprint1_acceptance.py:: test_tc10_same_brief_twice_creates_separate_instances	**Pass**
TC_11	US_02	Selection reason is shown	Brief submitted	1. Open the spawn result for a brief.	Each spawned agent shows why it was selected; each skipped agent shows why it was skipped.	Positive	backend/tests/test_sprint1_acceptance.py:: test_tc11_selection_reason_is_shown	**Pass**
TC_12	US_03	Table created on enrollment	Agent with a defined output schema	1. Enroll the agent. 2. Inspect the database.	A table exists with columns and data types matching the output schema.	Positive	backend/tests/test_sprint1_acceptance.py:: test_tc12_table_created_on_enrollment	**Pass**
TC_13	US_03	Unsafe table name	Backend running	1. Enroll an agent named with SQL text, for example x; DROP TABLE users.	Name is rejected or sanitised; no SQL runs and existing tables are untouched.	Negative	backend/tests/test_sprint1_acceptance.py:: test_tc13_unsafe_table_name_is_sanitised, test_tc13_unsafe_property_names_are_rejected	**Pass**
TC_14	US_03	Repeat table creation	Agent already enrolled with a table	1. Trigger table creation again for the same agent.	No error and no duplicate table; existing data is kept.	Positive	backend/tests/test_sprint1_acceptance.py:: test_tc14_repeat_table_creation_keeps_data, test_tc14_new_schema_fields_add_columns	**Pass**

TC_ID	US_ID	Test Scenario	Preconditions	Test Steps	Expected Result	Type	Automated Test	Status
TC_15	US_03	Table creation failure rolls back enrollment	Database set to fail on table creation	1. Enroll an agent while table creation fails.	Enrollment returns an error and no partial agent record remains in the registry.	Negative	backend/tests/test_sprint1_acceptance.py:: test_tc15_table_creation_failure_rolls_back_enrollment	**Pass**
TC_16	US_04	Reject a malformed schema	Backend running	1. Enroll an agent whose schema is invalid JSON or has an unknown type.	Enrollment is rejected with an error naming the problem field.	Negative	backend/tests/test_sprint1_acceptance.py:: test_tc16_invalid_json_is_rejected, test_tc16_unknown_ schema_type_is_rejected_naming_the_field, test_tc16_non_object_output_schema_is_rejected	**Pass**
TC_17	US_04	Store valid agent output	Enrolled agent with a table	1. Send output that matches the schema.	Output passes validation and is written to the agent's table.	Positive	backend/tests/test_sprint1_acceptance.py:: test_tc17_valid_output_is_written_to_the_agent_table	**Pass**
TC_18	US_04	Output with a missing required field	Enrolled agent with a table	1. Send output missing a required field.	Output is rejected, the step is marked Failed, nothing is stored, and the error is logged with agent ID and field.	Negative	backend/tests/test_sprint1_acceptance.py:: test_tc18_output_missing_a_required_field	**Pass**
TC_19	US_04	Output with the wrong data type	Enrolled agent with a table	1. Send a text value where a number is required.	Output is rejected and the error names the field and expected type.	Negative	backend/tests/test_sprint1_acceptance.py:: test_tc19_output_with_the_wrong_data_type	**Pass**
TC_20	US_05	Submit a brief from the UI	Frontend and backend running	1. Open /office. 2. Enter a multi-step brief. 3. Submit.	Spawned agents appear with status and selection reason.	Positive	frontend/e2e/sprint1.spec.ts "TC_20"	**Pass**
TC_21	US_05	Empty brief blocked in the UI	Frontend running	1. Click submit with an empty brief box.	Submission is blocked and a validation message is shown.	Negative	frontend/e2e/sprint1.spec.ts "TC_21"	**Pass**
TC_22	US_05	Agents page matches the API	Several agents enrolled	1. Open the agents page. 2. Compare with the list-agents API.	The same agents, capabilities and statuses are shown.	Positive	frontend/e2e/sprint1.spec.ts "TC_22"	**Pass**
TC_23	US_05	Backend error shown to the user	Backend returns a validation error	1. Trigger a rejected enrollment or brief from the UI.	A clear error message is shown and the page stays usable.	Negative	frontend/e2e/sprint1.spec.ts "TC_23"	**Pass**